

Labs

Write an action script to play and animation in flash (play and stop Animation)

Introduction

This lab demonstrates how to create a basic animation in Macromedia Flash and control it using ActionScript. Buttons are added to play and stop the animation, helping understand user interactivity and multimedia object behaviour in flash.

Objectives

- Create animation using flash.
- Use Actionscript to play and stop the animation.
- Understand multimedia object control using scripting.

Theory

Basic Multimedia Object (BMO):

A BMO is the smallest unit of multimedia content like an image, text or sound. It can be segmented (SG: BMO) with a fixed length (like a stored image) or streamed (ST: BMO) with no fixed duration (like live audio / video). BMOs form the foundation of all multimedia structures.

Simple Multimedia Object (SMO)

A SMO is formed by combining one or more BMOs along with header information. It includes metadata and data content and represents a simple multimedia unit like an image with audio. SMOs are used to structure and manage individual media components.

Compound Multimedia Object (CMO)

A CMO is a combination of several SMOs. It uses two headers: the orchestration header to manage timing, and the concatenation header to define layout and structure. CMOs allow complex multimedia like synced video, audio and pointer movements.

Abstract Programming

Abstract programming means writing code that focuses on logic rather than details. In Flash, this is done through ActionScript where you control animation (like play/stop) using simple scripts, without handling how the system processes it internally.

Functions in ActionScript

Function in Actionscript are reusable code blocks triggered by events like button clicks. For example, `on(release){stop();}` stops the animation when the stop button is pressed. These function make multimedia projects interactive and user-driven, enabling better control and flexibility.

Steps Involved

- 1) OpenFlash → New document (Ctrl+N)
- 2) Create layers: "Animation" and "Buttons".
- 3) Draw objects: Use Oval Tools (O), draw circle.
- 4) Convert to Symbol: Press F8 → Select "Movie Clip".
- 5) Create Motion Tween:
 - Insert keyframes (F6) at frame 1 and 30.
 - Move object in frame 30.
 - Right-Click → Create Motion Tween.
- 6) Add Buttons:
 - Draw rectangles → convert to buttons
 - Label as 'PLAY' and 'STOP'
- 7) Add ActionScript: Open Actions panel, write:
// STOP Button
on(release) { stop(); }
// PLAY Button
on(release) { play(); }
- 8) ~~Enter~~ Test Movie: Press ctrl + Enter

Shortcut Keys:

Action	Shortcut
New document	Ctrl+N
Convert to Symbol	F8
Insert keyframe	F6
Open Action panel	F9
Test Movie	Ctrl + Enter
Oval Tool	O
Selection Tool	V

Result

The animation was successfully created and controlled using 'PLAY' and 'STOP' buttons implemented with Actionscript.

Conclusion

This lab illustrated how Actionscript provides an abstract programming interface to control multimedia objects in Flash. Understanding BMO, SMO and concept CMO concepts aids in designing and managing interactive multimedia systems effectively.